

Injection Moulding - Decompression (Suck-Back) Setting Guide

1. What decompression (suck-back) is

Decompression, or suck-back, retracts the screw a small set distance after recovery to relieve pressure at the nozzle. It is set as a stroke (a few mm, scaled to screw diameter and resin).

2. Purpose and correct setting [p2]

Too little suck-back leaves nozzle pressure high and causes drool and stringing at the open nozzle. Too much suck-back pulls air back into the melt, causing splay, silver streaks and bubbles. Set the smallest suck-back that just stops drool, then confirm no air entrainment. Correct suck-back gives a clean, repeatable nozzle shot after shot.

3. Symptoms of a mis-set stroke

A mis-set decompression stroke MIMICS a worn non-return valve: the cushion can look erratic and shot weight can vary. Therefore always verify the decompression setting FIRST, before condemning and replacing the check ring / NRV.

4. Interaction with the cushion

Suck-back is measured from the cushion position. If the cushion itself is unstable because the NRV is leaking, the suck-back will look erratic even when the setting is correct. Stabilise / verify the cushion (see the cushion-stability reference) to tell a setting problem from a mechanical NRV problem.